

2.1 Basics of Algebra

Algebra is a branch of mathematics that uses letters and symbols to represent numbers. These letters allow us to describe relationships between quantities, develop general rules, and solve problems involving unknown values. For example,

$$x + 5 = 12$$

contains the letter x , which represents an unknown number. The objective is often to determine the value of the unknown. Algebra is used extensively throughout engineering, science and technology.

Definition

A letter used to represent a number is called a variable. Common variables include

$$x, \quad y, \quad z, \quad a, \quad b, \quad t$$

A variable may represent a quantity whose value is unknown or a quantity that can change.

2.1.1 Algebraic Terms

An algebraic expression is made up of one or more terms. For example,

$$3x + 5$$

contains two terms:

$$3x \quad \text{and} \quad 5$$

Similarly,

$$2a - 7b + 4$$

contains three terms:

$$2a, \quad -7b, \quad 4$$

Notice that the sign belongs to the term immediately to its right.

Definition

A term is a single quantity that is separated from other terms by addition or subtraction. For example, the expression

$$4x^2 - 3x + 7$$

contains the terms

2.1.2 Coefficients

The numerical part of a term is called its coefficient.

Worked Example

Consider the term

$$7x$$

The coefficient is

The variable is

Worked Example

Consider the term

$$-5y$$

The coefficient is

The variable is

If no coefficient is shown, it is understood to be 1. For example,

$$x = 1x$$

2.1.3 Like Terms

Only like terms can be combined. Like terms contain exactly the same variables raised to the same powers.

Worked Example

Simplify

$$3x + 2x$$

Both terms contain x , so they are like terms. Add the coefficients:

$$3 + 2 = 5$$

Therefore,

Worked Example

Simplify

$$7y - 4y$$

Subtract the coefficients:

$$7 - 4 = 3$$

Therefore,

However,

$$3x + 2y$$

cannot be simplified because x and y are different variables.

Definition

Like terms have identical variable parts. Examples:

$$3x, \quad -5x, \quad 8x$$

are like terms.

However,

$$3x, \quad 3y, \quad 3x^2$$

are not like terms.

2.1.4 Collecting Like Terms

Collecting like terms is one of the most common algebraic operations.

Worked Example

Simplify

$$4x + 7 + 3x - 2$$

Collect the x -terms:

Collect the constants:

Therefore,

Worked Example

Simplify

$$5a - 8 + 2a + 3$$

Collect like terms and simplify:

$$5a - 8 + 2a + 3 =$$

2.1.5 Expanding Brackets

When a bracket is multiplied by a quantity, every term inside the bracket must be multiplied.

Worked Example

Expand

$$3(x + 4)$$

Multiply every term inside the bracket by 3:

$$3(x + 4) =$$

Worked Example

Expand

$$2(4x - 5)$$

Multiply each term by 2:

$$2(4x - 5) =$$

Negative signs require particular care.

Worked Example

Expand

$$-7(x + 3)$$

The negative sign means multiply every term by -7 . Therefore,

$$-7(x + 3) =$$

Worked Example

Expand

$$-(2x - 7)$$

Multiply each term by -1 :

$$\begin{aligned} -(2x - 7) &= \\ &= -2x + 7 \end{aligned}$$

2.1.6 Checking Algebra

A useful way to check algebraic manipulation is to substitute a value. Consider

$$3(x + 4) = 3x + 12$$

Let

$$x = 2$$

The left-hand side becomes

$$3(2 + 4) = 18$$

The right-hand side becomes

$$3(2) + 12 = 18$$

Since both sides give the same result, the expansion is correct.

2.1.7 Common Mistakes

Common algebra mistakes include:

- Combining unlike terms.
- Forgetting that signs belong to terms.
- Multiplying only the first term when expanding brackets.
- Losing negative signs.
- Assuming $3x + 2y = 5xy$.

2.1.8 Engineering Application

Engineering Application

Algebra allows engineers to describe relationships between quantities before numerical values are known.

For example, Newton's Second Law states that

$$F = ma$$

where: F is force, m is mass and a is acceleration.

The formula is valid for any object, regardless of its mass or acceleration.

Algebra therefore allows engineers to develop general equations that can later be applied to specific situations.

2.1.9 Exercises

Exercise

1. State the coefficient of:

- (a) $7x$
- (b) $-4y$
- (c) $12a$
- (d) x

2. Simplify:

- (a) $3x + 5x$
- (b) $8y - 3y$
- (c) $6a + 4 - 2a + 5$
- (d) $7m - 8 + 2m - 3$

3. Expand:

- (a) $2(x + 5)$
- (b) $3(a - 2)$
- (c) $4(2x + 3)$
- (d) $5(3y - 1)$

4. Expand:

- (a) $-(x + 4)$
- (b) $-(2a - 3)$
- (c) $-3(x - 2)$
- (d) $-2(4y + 1)$

5. Simplify:

- (a) $3x + 2 + 4x - 5$
- (b) $5a - 7 + 3a + 2$
- (c) $2(3x + 4)$
- (d) $4(2y - 3) + 5y$

Solutions to Exercise 2.1.9

- 1. (a) 7
- (b) -4

- (c) 12
 - (d) 1
2. (a) $8x$
- (b) $5y$
 - (c) $4a + 9$
 - (d) $9m - 11$
3. (a) $2x + 10$
- (b) $3a - 6$
 - (c) $8x + 12$
 - (d) $15y - 5$
4. (a) $-x - 4$
- (b) $-2a + 3$
 - (c) $-3x + 6$
 - (d) $-8y - 2$
5. (a) $7x - 3$
- (b) $8a - 5$
 - (c) $6x + 8$
 - (d) $13y - 12$